

Fantasy Football Tracker — technical reference

Personal tracker: **Python collector** POSTs snapshots into **Spring Boot 3.5 + MySQL**, and **Angular 21** GETs a dashboard. The frontend never writes. Gameweeks are **rows**, not tables.

Sources

source	Slug example	How data arrives
sofascore	premier-league, laliga, ...	Logged-in SofaScore Fantasy XHRs + <code>SOFASCORE_SESSION</code>
fpl	premier-league-fantasy	Public FPL JSON (<code>FPL_ENTRY_ID</code>)
laliga-fantasy	laliga-fantasy-oficial	Excel workbook (<code>import-excel</code>)
wsl	wsl-fantasy	Official my-team XHR (<code>x-game-token</code>) + public feeds

Constraints that shaped the code: collector POSTs / frontend only GETs; MySQL on 3307; never commit `.env`, cookies, or tokens; do not invent SofaScore API paths or scrape HTML; do not reverse official LaLiga/WSL apps; one process on port 8080.

Architecture

```
collector (pull / import / import-excel)
  POST /api/v1/ingest/snapshots
  POST /api/v1/ingest/transfers
    → IngestService upserts entities
frontend GET /api/v1/competitions...
  → CompetitionQueryService + branding/crests/portraits
```

Captain **display** is always $\times 2$ ($\times 3$ if triple captain). Ingest stores **raw** points. Official WSL `totalPoints` already includes $\times 2$, so the collector divides the captain's value by 2 before POST.

Backend (Spring Boot)

Package: `com.fantasytracker.backend`. Profile `mysql` uses XAMPP MariaDB on 3307 (`fantasy_tracker`). Default profile is H2 + seed JSON for tests.

BackendApplication

- `main` — boots Spring. Created as the single process that owns port 8080.

Config

WebConfig.addCorsMappings

Allows the Angular origin (`fantasy.cors.allowed-origins`, default `http://localhost:4200`) on `/api/**`. Exists so the dashboard can GET localhost:8080 from localhost:4200.

SeedRunner.run

`ApplicationRunner` (on unless `fantasy.seed.enabled=false`). If the competition table is empty, ingests `seed/pl-gw1.json` and `seed/pl-gw2.json` so H2/demo boots with "Sofa Saints". Skipped on a populated MySQL database.

HTTP — ingest (`IngestController`)

Optional header `X-Ingest-Token` when `fantasy.ingest.token` is set.

Method	Path	Why
<code>ingest</code>	POST <code>/api/v1/ingest/snapshots</code>	Collector (and Excel/FPL/WSL adapters) upsert one gameweek squad + scores.
<code>ingestTransfers</code>	POST <code>/api/v1/ingest/transfers</code>	Official transfer rows (paired in/out, or unpaired LaLiga buys/sells).
<code>cleanupSeed</code>	POST <code>/api/v1/ingest/cleanup-seed</code>	Removes players whose <code>externalId</code> starts with <code>seed-</code> after switching to live data.

HTTP — read (`CompetitionController`)

Frontend-only surface.

Method	Path	Why
<code>list</code>	GET <code>/api/v1/competitions</code>	Dropdown: name, season, branding, gameweek numbers.
<code>team</code>	GET <code>/api/v1/competitions/{id}/team?gameweek=</code>	Squad for a week (latest if omitted).
<code>gameweek</code>	GET <code>/api/v1/competitions/{id}/gameweeks/{n}</code>	Same payload keyed by week number.
<code>compare</code>	GET <code>/api/v1/competitions/{id}/compare?from=&to=</code>	Week-vs-week player deltas.
<code>totals</code>	GET <code>/api/v1/competitions/{id}/totals</code>	Season sum of team week scores.

IngestService

Why it exists: one upsert path for SofaScore, FPL, WSL, and Excel so the dashboard schema stays stable.

- `upsertSnapshot` — competition, gameweek, team, picks, player scores; drops stale picks for that week; team points from the request or from starter points after `CaptainScoring.effective`.

- **upsertTransfers** — replaces that week's transfer rows; applies **transferPenalty** onto **TeamGameweekScore**.
- **removeSeedPlayers** — cleanup for **seed-*** ids.
- **findAllCompetitions** — used by tests/tooling.
- **upsertCompetition** / **upsertGameweek** / **upsertTeam** / **upsertPlayer** / **upsertPick** / **upsertPlayerScore** / **upsertTeamScore** — idempotent writes keyed by source+external id (or team+player+week).
- **upsertGameweekForTransfers** / **applyTransferPenalty** — transfers can arrive for a week that has no squad yet.
- **removeStalePicks** — a new XI replaces last week's leftover rows.
- **toJson** — stores pick **breakdown** (minutes, goals, ...) as JSON text.

CompetitionQueryService

Why it exists: GET payloads include derived media (crest, portrait, colours) and captain **display** points without the frontend knowing chips.

- **listCompetitions** — all competitions with branding + distinct gameweek numbers.
- **teamView** / **gameweekView** — same builder; **teamView** picks latest week if unspecified.
- **compare** — players in either week; points use captain multiplier per week's triple-captain flag.
- **totals** — ordered week scores + season total.
- **buildTeamView** — starters/bench, portraits, crests, transfers, transfer-market summary.
- **buildTransfers** — stored **gameweek_transfer** rows if present; otherwise **squad-diff** against the previous week (WSL and gaps in official feeds).
- **toTransfer (overloads)** — maps a player + price + counterpart into the dashboard transfer card.
- **buildTransferMarket** / **summarizeMarket** / **toCounterpartGroups** / **Accumulator** — LaLiga market vs release-clause totals (week + season).
- **previousGameweek** / **tripleCaptain** / **pickComparator** / **positionOrder** — GK→DEF→MID→FWD, starters before bench.
- **indexPicks** / **indexScores** / **pointsOf** / **resolveGameweek** / **requireCompetition** / **requireTeam** / **requireGameweek** — lookups that 404 as 404-style failures.

Scoring and media helpers

CaptainScoring

- **multiplier** — 1, 2, or 3.
- **effective** — raw × multiplier. Created so SofaScore/FPL/WSL all store raw and display chips the same way.

CompetitionBranding

Maps **source** + **externalId** onto SofaScore unique-tournament ids (17 PL, 8 LaLiga, 23 Serie A, 34 Ligue 1, 35 Bundesliga, 7 UCL, 679 UEL, 242 MLS, 325 Brasileirão, **1044 WSL**). Official FPL/LaLiga/WSL reuse those ids even though **source** is not **sofascore**.

- **logoUrl** / **logoDarkUrl** / **flagUrl** / **countryName** / **primaryColor** / **secondaryColor**
- **brandingTournamentId** / **numericId** / **country** / **colors** — private mapping.

ClubCrests.url

FPL uses Premier League badge CDN `t{code}.png`. Everyone else uses `img.sofascore.com/api/v1/team/{id}/image`, with name fallbacks for PL/LaLiga clubs when the id is missing. WSL is **not** mapped through men's PL names (Chelsea Women must keep a SofaScore women's team id).

PlayerPortraits.url

Numeric SofaScore player id → `img.sofascore.com/api/v1/player/{id}/image`. Non-numeric ids (unresolved WSL `ws1-...`, FPL `fp1-...`) get no portrait.

TransferChannels

- **isMarket / channel** — blank/"Market" vs any other counterpart = release clause. Exists for official LaLiga Excel deals.

Entities (JPA)

Lombok `@Data`. Tables match `sql/schema.sql`.

Entity	Role
Competition	Unique (<code>source</code> , <code>externalId</code> , <code>season</code>).
Gameweek	Week row (<code>number</code> , <code>status</code> , <code>dates</code>).
FantasyTeam	One team per competition.
Player	Unique (<code>source</code> , <code>externalId</code>); <code>externalId</code> is the SofaScore player id when resolved.
SquadPick	Role, captain/VC, price, injured flag.
PlayerGameweekScore	Raw <code>points</code> , optional <code>rating</code> + <code>breakdown</code> .
TeamGameweekScore	Week total, triple-captain flag, transfer penalty.
GameweekTransfer	Nullable in/out players, prices, counterpart, channel, sort order.

Repositories

Spring Data query methods — no custom SQL. They exist so ingest and GET stay indexed on natural keys.

- `CompetitionRepository.findBySourceAndExternalIdAndSeason`
- `GameweekRepository.findByCompetition_IdAndNumber`,
`findByCompetition_IdOrderByNumberAsc`
- `FantasyTeamRepository.findByCompetition_Id`
- `PlayerRepository.findBySourceAndExternalId`, `findByExternalIdStartingWith` (seed cleanup)
- `SquadPickRepository.findByFantasyTeam_IdAndGameweek_Id`,
`findByFantasyTeam_IdAndPlayer_IdAndGameweek_Id`, `deleteByPlayer_Id`

- `PlayerGameweekScoreRepository.findByPlayer_IdAndGameweek_Id, findByGameweek_IdAndPlayer_IdIn, deleteByPlayer_Id`
- `TeamGameweekScoreRepository.findByFantasyTeam_IdAndGameweek_Id, findByFantasyTeam_IdOrderByGameweek_NumberAsc`
- `GameweekTransferRepository.findByFantasyTeam_IdAndGameweek_IdOrderBySortOrderAsc, findByFantasyTeam_IdOrderByGameweek_NumberAscSortOrderAsc, deleteByFantasyTeam_IdAndGameweek_Id`

DTOs (records)

- `SnapshotRequest` (+ nested competition/gameweek/team/player/pick) — ingest body.
- `TransfersRequest` (+ round + pair) — ingest transfers; in/out may be null for LaLiga.
- `CompetitionResponse` — list item including branding URLs/colours.
- `TeamViewResponse` — dashboard squad: picks, transfers, transfer-market summary, captain display fields (`points`, `basePoints`, `captainMultiplier`).
- `CompareResponse` / `PlayerDelta`
- `TotalsResponse` / `GameweekTotal`

Tests

- `ApiSliceTest` — MockMvc slice: SofaScore chips, LaLiga market, FPL badges, WSL branding (tournament 1044), captain display.
- `BackendApplicationTests` — context load.

Frontend (Angular 21)

Standalone components, signals, `HttpClient`. One route: the dashboard.

Bootstrap

- `main.ts` — `bootstrapApplication(App, appConfig)`.
- `appConfig` — router, HTTP, global error listeners.
- `routes` — `'` → `DashboardComponent`; wildcard redirects home.
- `App` — shell with `<router-outlet>`. No logic; the dashboard is the product.

CompetitionService

Thin GET client at `http://localhost:8080/api/v1/competitions`. Created so the UI never POSTs.

- `getCompetitions`
- `getTeam(id, gameweek?)`
- `getGameweek(id, n)` — available but dashboard uses `getTeam`.
- `compare` / `totals`

tracker.ts models

TypeScript mirrors of GET JSON: `Competition`, `TeamView`, `PickView`, `TransferView`, `TransferMarket*`, `CompareView`, `PlayerDelta`, `TotalsView`. Exist so templates stay typed.

DashboardComponent

Why: one page for every league — branding, XI, bench/squad, transfers, compare, totals.

Signals: `competitions`, `selectedId`, `gameweek`, `fromGw`, `toGw`, `teamView`, `compareView`, `totalsView`, `error`, `loading`.

Computed

- `selectedCompetition` — current dropdown row.
- `pageBrand` — CSS variables from API primary/secondary + logo (page background).
- `starters` / `bench` — split on `role === 'starter'` (LaLiga “squad” still uses non-starter role).
- `isOfficialLaLiga` / `isOfficialFpl` — copy and FPL neon classes (`#01fc84`, `#39a1f9`, `#8c46ff` mixed with PL purple/pink).
- `reserveHeading` — “Squad” vs “Bench”.
- `showTransferCounterpart` — LaLiga or any move with a counterpart.

Methods

- `constructor` — loads competitions; selects the first.
- `selectCompetition` — latest week, compare from first→latest, refresh all three GETs.
- `onCompetitionChange` / `onGameweekChange` / `onCompareChange` — template bindings.
- `transferLabel` / `priceFormat` / `counterpartLabel` / `hasMarketDeals` — LaLiga market wording and 6-decimal prices (Purić-style).
- `hideImage` — hide broken crest/portrait (no alt junk).
- `signed` — `+n` / `n` for compare deltas.
- `refreshAll` / `loadTeam` / `loadCompare` / `loadTotals` — parallel-ish HTTP; errors set `error`.

Templates/styles: `dashboard.html` (branded page, FPL extra classes), `dashboard.css` (league colours + FPL neon overlay).

Python collector

Package `collector`, CLI `python -m collector`. Loads repo-root then `collector/.env`. Raw JSON goes to `collector/cache/<slug>/` (gitignored).

CLI (`__main__.py`)

- `main` — subcommands `import`, `import-excel`, `sync`, `pull`.
- `_run_import` — local JSON (squad or transfers export).
- `_run_import_excel` — official LaLiga workbook.
- `_run_sync` — legacy single-GW SofaScore URL.
- `_run_pull_command` — every SofaScore target in `.env`, plus FPL if `FPL_ENTRY_ID`, plus WSL if gameplay id + game token. `--competition` selects one slug (`premier-league-fantasy`, `ws1-fantasy`, ...).

Canonical models (`models.py`)

- **PlayerPayload / PickPayload / Snapshot** — POST `/ingest/snapshots`. `Snapshot.to_dict` omits null team points.
- **TransferPlayer / TransferPair / TransferRound / TransfersBatch** — POST `/ingest/transfers`.
- **FantasyAdapter** — old interface (`fetch_competitions / fetch_squad / fetch_gameweek_scores`); SofaScore and file import still implement it. FPL/WSL use dedicated `pull_*` functions instead.

BackendPublisher

- **publish / publish_transfers** — POST with optional `X-Ingest-Token`. `BACKEND_URL` defaults to `http://localhost:8080`.

Pull orchestration (`pull.py`)

- **PullTarget** — slug + competition/squad/transfers/gameweek URLs.
- **pull_targets_from_env** — Premier League unprefixed `SOFASCORE_*` or `SOFASCORE_PREMIER_LEAGUE_*`; other leagues `SOFASCORE_LALIGA_*`, `SERIE_A`, `LIGUE_1`, `BUNDESLIGA`, `CHAMPIONS_LEAGUE`, `EUROPA_LEAGUE`, `MLS`, `BRASILEIRAO`.
- **complete_target** — from a Fantasy `.../competition/{id}` URL, derive `/transfers` and `/round/{roundId}/squad`.
- **_target_from_prefix** — env → `PullTarget`.
- **adapter_for_target** — `SofaScoreAdapter` for that target.
- **run_pull** — meta + rounds → `currentRound.id` (or `--gameweek` sequence) → squad → injury overlay → optional transfers → POST (unless `--dry-run`).
- **_save_json** — cache dump.

SofaScore Fantasy adapter (`adapters/sofascore.py`)

Why: replay **your** Network JSON URLs with your cookie; impersonate Chrome (`curl_cffi`) because Python `requests` was WAF 403.

- **normalize_position**
- **snapshot_from_payload** — canonical snapshot, SofaScore squad, or “sofascore-like” shapes.
- **is_transfers_export / transfers_from_payload**
- **_from_canonical / _from_sofascore_squad / _from_sofascore_like**
- **_injured_from_sofascore / _first_unique_tournament / _normalize_season / _date_from_timestamp**
- **_transfer_player / _club_external_id / _optional_float**
- **SofaScoreSessionError** — 401 → refresh `SOFASCORE_SESSION`.
- **SofaScoreAdapter**
 - **fetch_competitions / fetch_meta / fetch_rounds / fetch_squad / fetch_transfers / fetch_gameweek_scores**
 - **_get_json / _require_round_id**
- **round_id_from_meta** — current/next/previous or `userRounds` calendar by sequence.
- **_browser_headers / _http_get / _env_or / _normalize_session_cookie / _format_url** — strip Cookie: prefix; rewrite `/round/{digits}/`.
- **load_snapshot_file**

Round overlay (`adapters/sofascore_round.py`)

Uses **known** SofaScore paths only: `unique-tournament/{id}/seasons`, `.../season/{id}/players`, `.../events/round/{n}`, `event/{id}/lineups`, `team/{id}/players`.

- `current_season_id` — first season in the list (current).
- `overlay_from_lineups` — ratings + injured from `missingPlayers` (not doubtful).
- `fetch_round_overlay` — all events in a round.
- `fetch_injured_player_ids` — live `injury.status == out`.
- `apply_round_overlay / annotate_snapshot_injuries`
- `_injured_for_team / _is_injured_missing / _player_id / _optional_int`

Player directory (`adapters/sofascore_directory.py`)

Why: official FPL/WSL ids are not SofaScore ids; portraits need numeric player ids.

- `fold_name / club_key` — accents, FC/Women/Lionesses, Brighton & Hove, Man Utd aliases.
- `fetch_tournament_players / fetch_premier_league_players`
- `resolve_sofascore_player` — club then name tokens.
- `search_sofascore_player` — `search/all` fallback (same family as LaLiga id fill).
- `_tokens_match`

Tournament constants: PL **17**, WSL **1044**, WSL2 **10553**. Unique-tournament **44** is 2. Bundesliga, not WSL.

Official FPL (`adapters/fpl.py`)

Public `fantasy.premierleague.com/api`. No cookie.

- `fpl_entry_id / pull_fpl` — bootstrap, entry, history, transfers, per-GW picks + live; SofaScore overlay on tournament 17.
- `catalog_from_bootstrap / competition_payload / team_payload / snapshot_from_picks / transfers_from_fpl`
- `_pick_from_fpl / _transfer_pair / _transfer_player / _player_from_element / _sofascore_player_id`
- `_safe_round_overlay / _current_injured_ids` — FPL status `i` + SofaScore missing/out.
- `_event_status / _season / _price` — prices in tenths of a million.
- `_optional_float / _get_json / _save_json`

Captain live points are **raw**; dashboard $\times 2$.

Official WSL (`adapters/wsl.py`)

Auth is request header `x-game-token` (env `WSL_GAME_TOKEN`, fallback `WSL_BEARER`). Auth0 Bearer is identity, not my-team. Public config/feeds discovered from official `configurations.json` / mixApi (not invented gameplay paths).

- `wsl_game_token / wsl_gameplay_id / wsl_configured / pull_wsl`
- `competition_payload / team_payload / snapshot_from_my_team / transfers_from_squads` — transfers are week-to-week `arrTeam` diffs (no official transfers XHR).
- `_pick_from_wsl / _player_from_row / _sofascore_player_id / _transfer_player`

- `_current_injured_ids` / `_safe_round_overlay` / `_safe_wsl_players` — overlay 1044 + 10553.
- `_gameweek_status` / `_starts_at` / `_season`
- `_raw_points` — my-team `totalPoints` **already includes captain ×2**; store half so the API does not double again.
- `_team_points` / `_position_name` / `_short_id` / `_index` / `_optional_float` / `_feed_value` / `_get_json` / `_save_json`

Feeds used: `/fantasy/services/gameplay/{id}/{week}/my-team`, `/feeds/tour/details/1.json`, `/feeds/players/matchday_en_1_{week}.json`, `/feeds/fixtures/fixtures_en_1.json`.

Official LaLiga Excel (`adapters/excel_laliga.py`)

App-only game (100M€ market, XI + unused squad, unpaired buys/sells, no captains).

- `load_workbook_payloads` — snapshots + transfer batch.
- `_competition_and_team` / `_snapshots_from_squads` / `_transfers_from_sheet`
- `_pick_from_row` / `_deal_from_row` / `_transfer_player`
- `_key_values` / `_table_rows` / `_id_or_slug` / `_slug` / `_digits` / `_int` / `_optional_float` / `_optional_text` / `_position` / `_action` / `_injured` / `_role`

Yellow EXAMPLE rows are skipped. Do not overwrite the empty template with personal GW data.

File import (`adapters/file_import.py`)

- `FileImportAdapter.fetch_competitions` / `fetch_squad` / `fetch_gameweek_scores` / `_load` — saved Network JSON when URLs are unavailable.

Excel template builder (`templates/build_laliga_fantasy_workbook.py`)

- `build` — writes `laliga-fantasy-oficial.xlsx`.
- `_readme` / `_competition` / `_squads` / `_transfers` / `_clubs` / `_header_row` / `_fill_row` / `_list` — styling and dropdowns so the importer has a stable schema.

Collector tests

`test_fpl`, `test_wsl`, `test_pull`, `test_sofascore_*`, `test_excel_laliga`, `test_sofascore_directory` — mapping and env targeting without printing secrets.

Environment (gitignored)

Variable	Purpose
<code>SOFASCORE_SESSION</code>	Cookie value from a Fantasy XHR
<code>SOFASCORE_*</code> / <code>SOFASCORE_<LEAGUE>_*</code>	Competition / squad / transfers URLs
<code>FPL_ENTRY_ID</code>	Public FPL team id
<code>WSL_GAMEPLAY_ID</code>	Gameplay UUID in my-team path
<code>WSL_GAME_TOKEN</code> (or <code>WSL_BEARER</code>)	<code>x-game-token</code> header (~24h). Not Auth0.

Variable	Purpose
BACKEND_URL / INGEST_TOKEN	Publisher target

Never log tokens. HTTP 401 on SofaScore → refresh cookie. HTTP 401 on WSL → copy a fresh `x-game-token`.